

4 a) Installation of Citrix XenServer 4 b) Live Migration of Virtual Machines using Xen Center

Aim:

To install and configure Citrix XenServer for virtualization of multiple operating systems , use Xen Center for graphical management of virtual machines and perform live migration of VMs between physical hosts without downtime.

Xen

Xen is a Type 1 (bare-metal) hypervisor that enables the execution of multiple virtual machines (VMs) on a single physical machine by abstracting the hardware. It supports both paravirtualization and hardware-assisted full virtualization (via Intel VT or AMD-V).

Virtualization Type: Para or Full?

Feature	Xen
Supports	Both Paravirtualization (PV) and Hardware-assisted Full Virtualization (HVM)
Default Mode	Originally designed for Paravirtualization (efficient, faster)
With Hardware Assist	Uses HVM mode for OSes like Windows (needs CPU virtualization support)

Xen Type 1 vs Type 2 Hypervisors

Feature	Type 1 (Xen)	Type 2
Runs on	Bare metal (hardware directly)	Runs on host OS
Examples	Xen, VMware ESXi, Microsoft Hyper-V	VirtualBox, VMware Workstation
Performance	High performance, low overhead	Slightly lower performance
Security	More secure (minimal layers)	Depends on host OS security

Equivalent Hypervisors to Xen

Hypervisor	Type	Notes
KVM	Type 1 (in Linux kernel)	Integrated with Linux, uses hardware virtualization
VMware ESXi	Type 1	Enterprise-grade, GUI support
Microsoft Hyper-V	Type 1	Windows-based hypervisor
VirtualBox	Type 2	Desktop virtualization

Requirements for Live Migration in Xen

- Shared Storage (e.g., NFS, iSCSI, SAN): Both hosts must access the same virtual disk.
- Same CPU family: Compatible CPU features between source and destination.
- Network Connectivity: Low-latency, high-speed network between hosts.
- Xen-compatible tools: xm, xl, or XenAPI-based management like XenCenter.
- Proper Configuration: Dom0 should have migration tools enabled, with firewall ports open.

XenServer vs XenCenter

Feature	XenServer	XenCenter
Definition	Commercial distribution of Xen hypervisor	GUI-based management tool for XenServer
Runs on	Physical Server (bare-metal)	Windows desktop (client-side software)
Function	Creates and runs VMs	Manages, monitors, and controls XenServer
Open Source?	XenServer is partly open-source (Citrix)	XenCenter is free but not open-source

a)

Citrix XenServer is a virtualization platform based on the open-source Xen Hypervisor. It enables efficient creation, management, and execution of multiple virtual machines (VMs) on a single physical host.

Installation Steps:

1. Download ISO:

- Get the official XenServer ISO from Citrix website.

2. Create Bootable Media:

- Use tools like Rufus or Ventoy to create a bootable USB with XenServer ISO.

3. BIOS Settings:

- Enable VT-x / AMD-V (hardware virtualization) and set USB as first boot device.

4. Boot and Install:

- Insert the USB and boot the server.
- Follow the on-screen instructions:
 - Select installation drive (e.g., local disk).

- Configure network settings (static IP recommended).
- Set root password.
- Choose to install XenServer tools

5. Complete Installation:

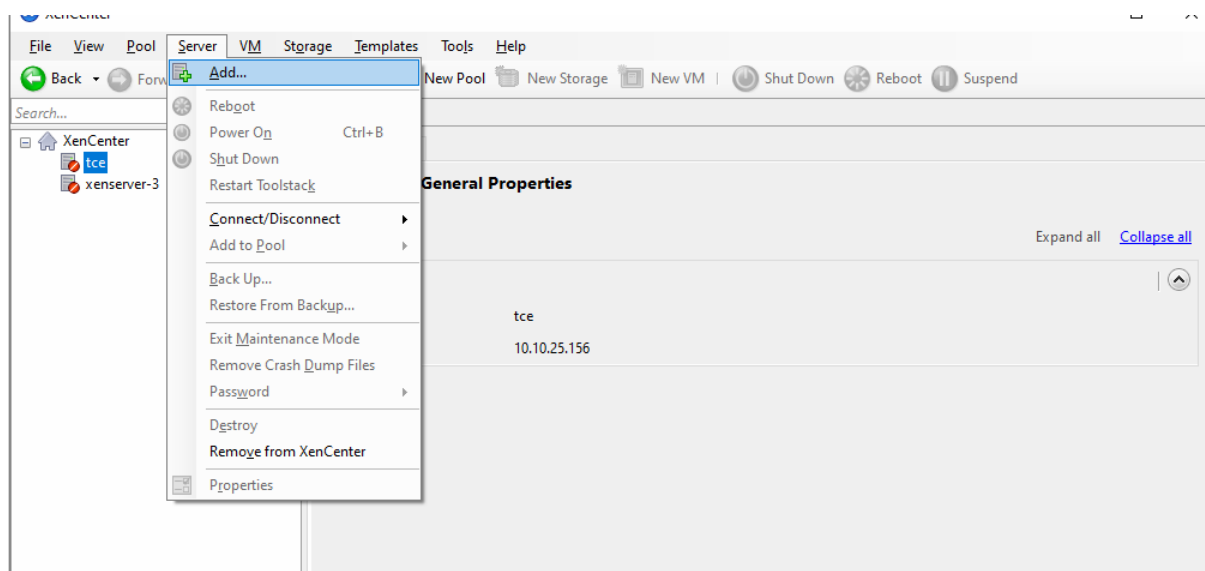
- Reboot system after installation.
- XenServer boots with a console UI showing IP address and system info.

4b) Live Migration of Virtual Machines using XenCenter

Live Migration refers to moving a running virtual machine from one physical host to another without shutting it down, using XenCenter's GUI.

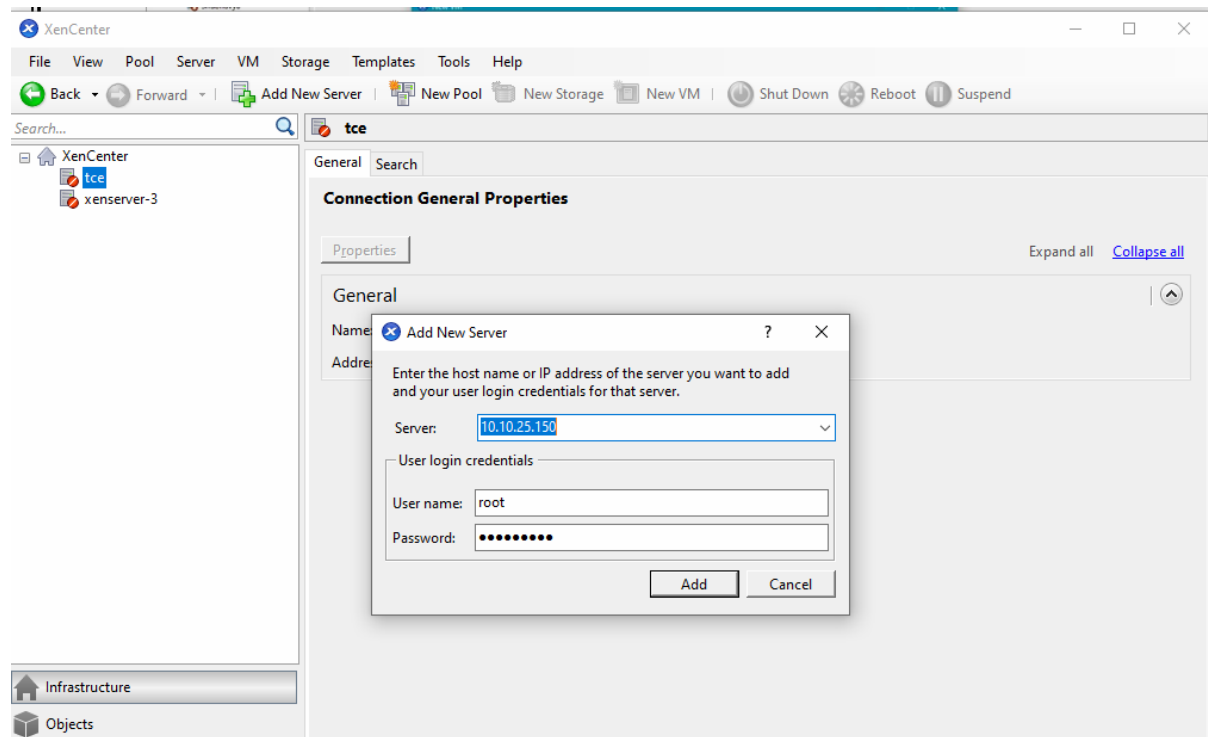
Steps 1:

Add the new server by clicking the server in the xencenter and server-> Add



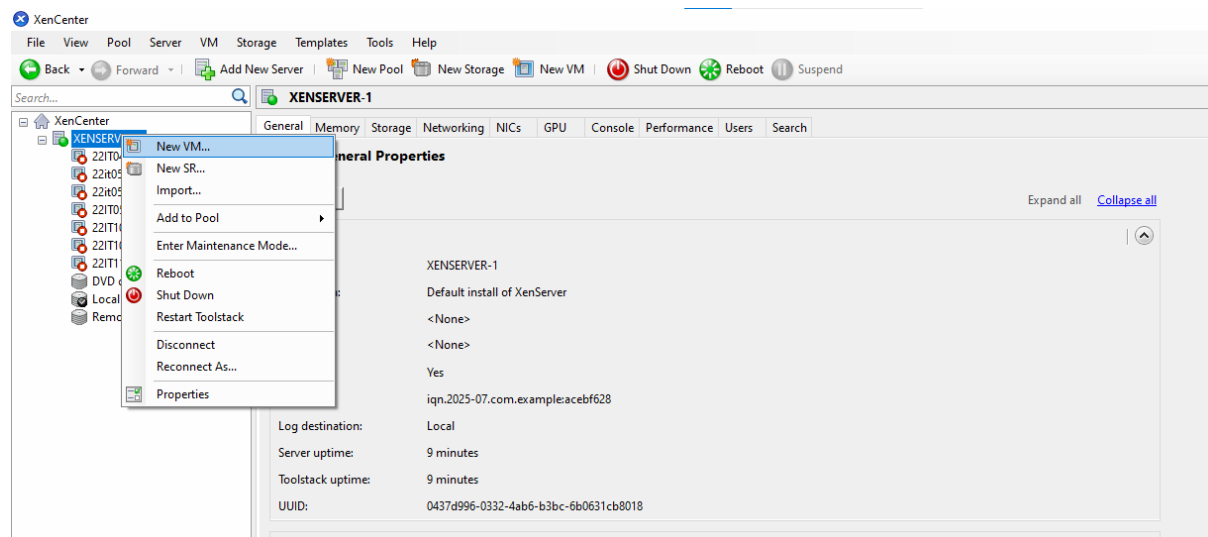
Steps 2:

Add the xenserver 1 by entering the Server , Username and password and click add.



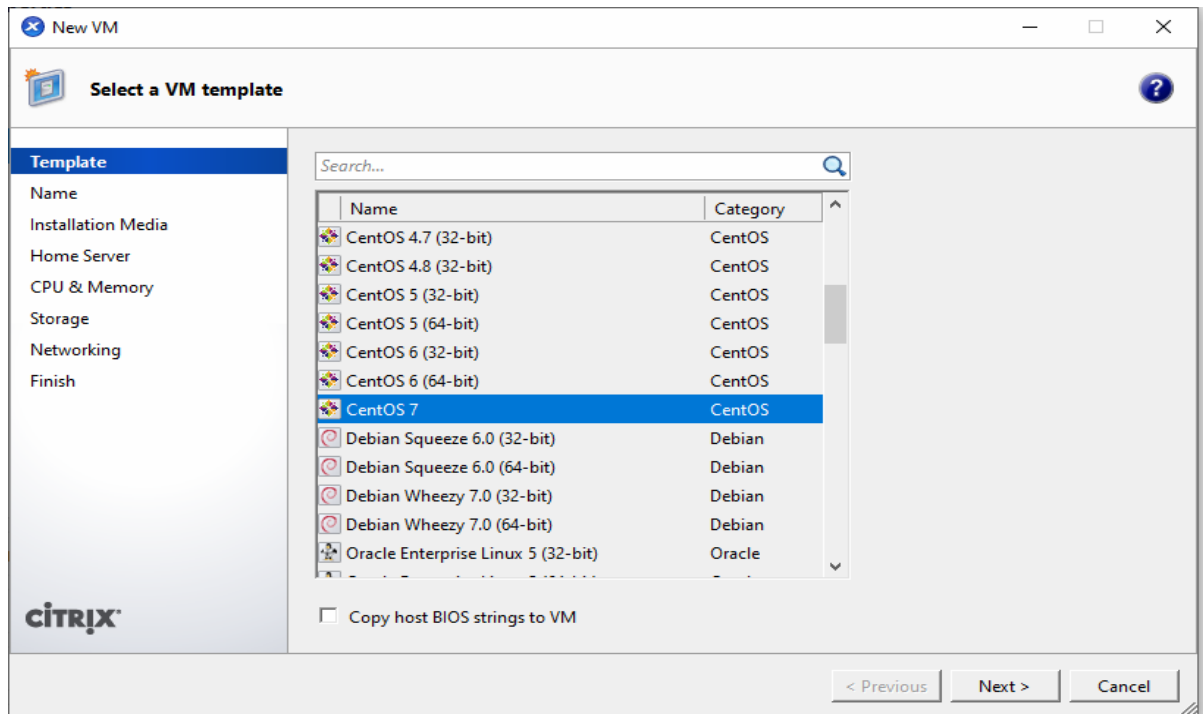
Steps 3:

Next add the vm of yours by clicking the xenserver 1 (right click) and add click new vm

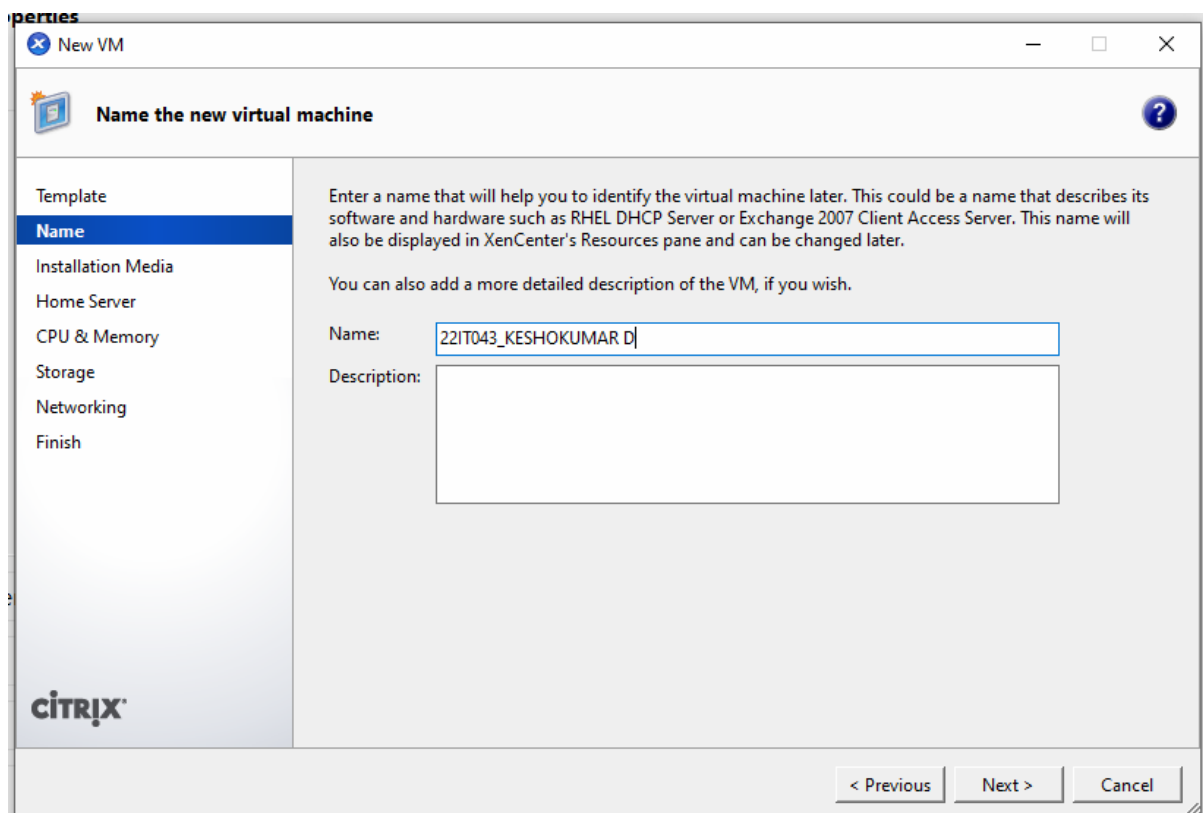


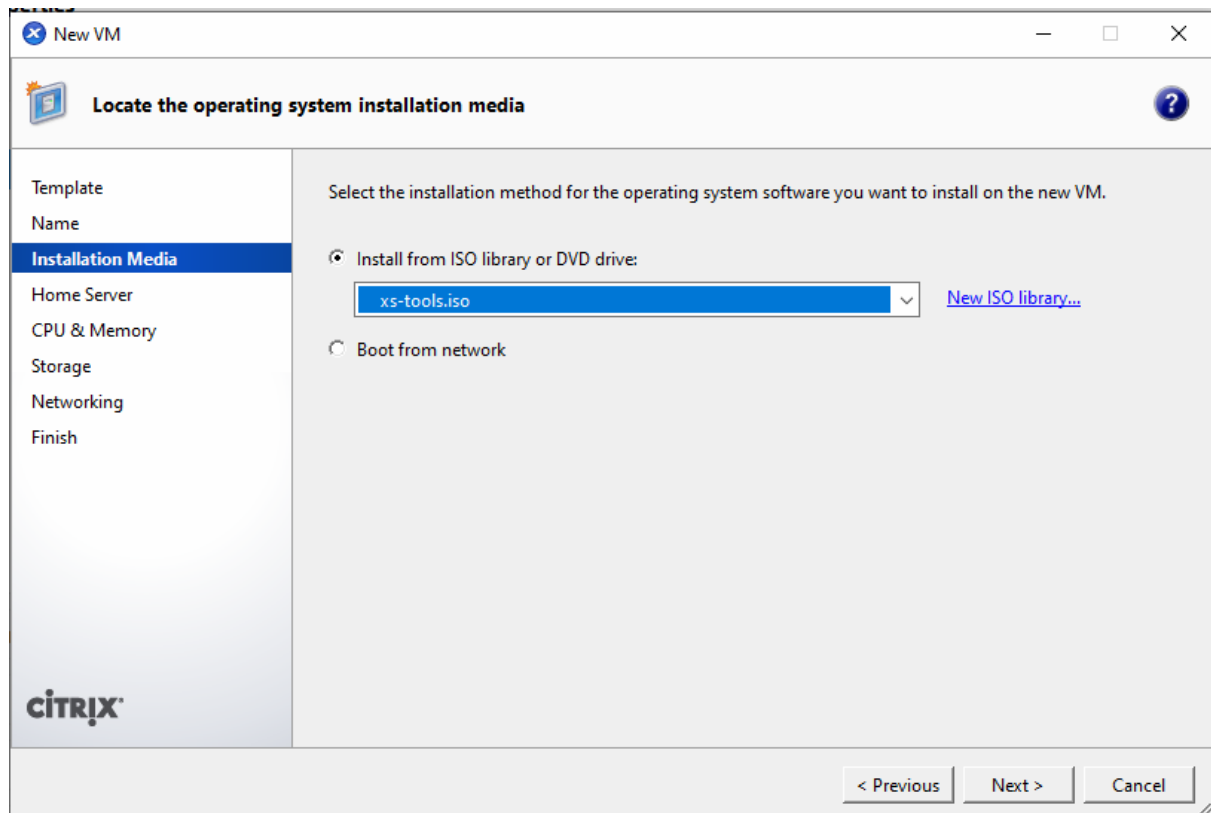
Steps 4:

Click the template example citric or ubunttu os or cent os(os which you have).

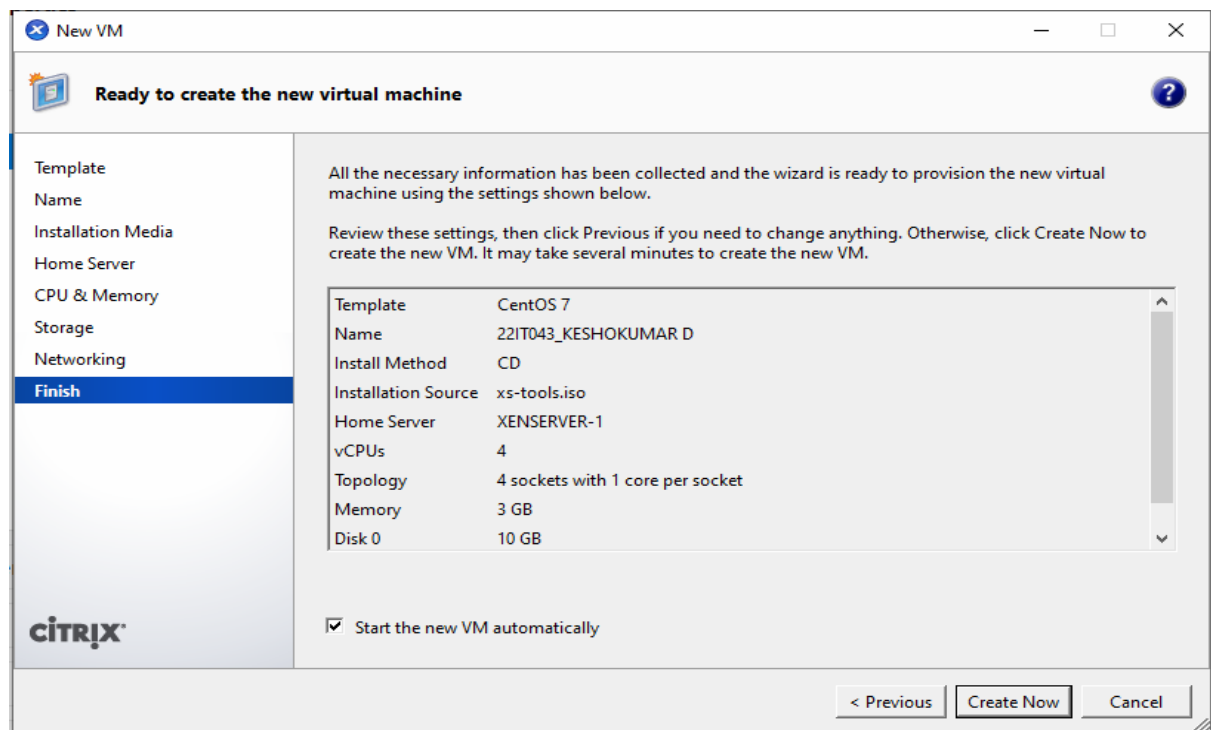


Steps 5: Give the name for your Vm and click next.

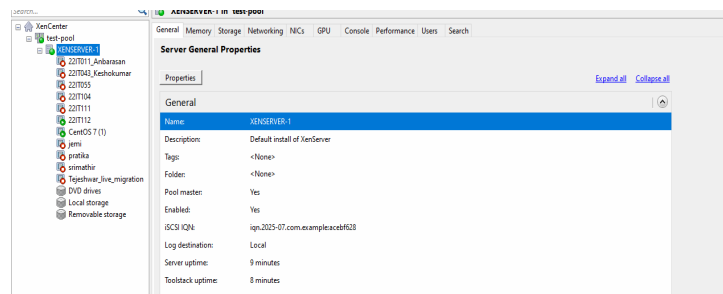


Steps 6: Select the iso file inside and click the next**Steps 7:**

Finally click the create now option then your vm has been created successfully.

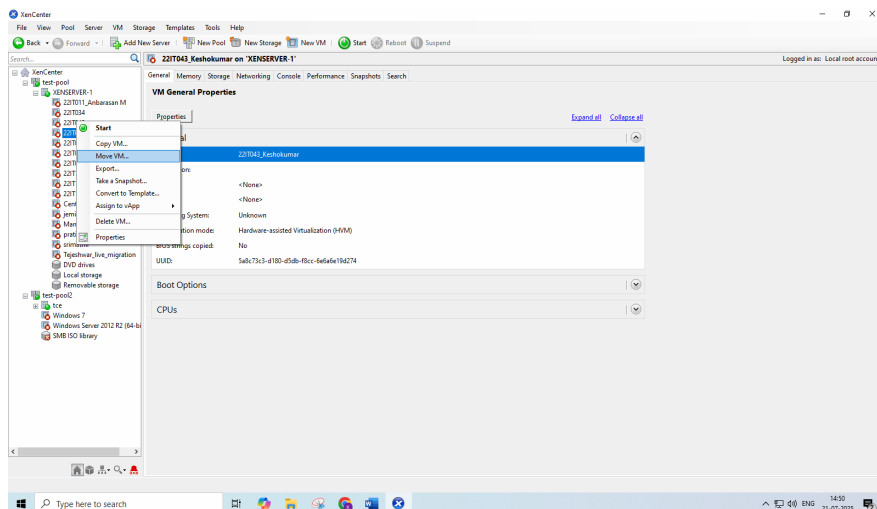


Steps 8: Create a xenserver 2 same procedure followed in first xenserver1 and click the xenserver 1.

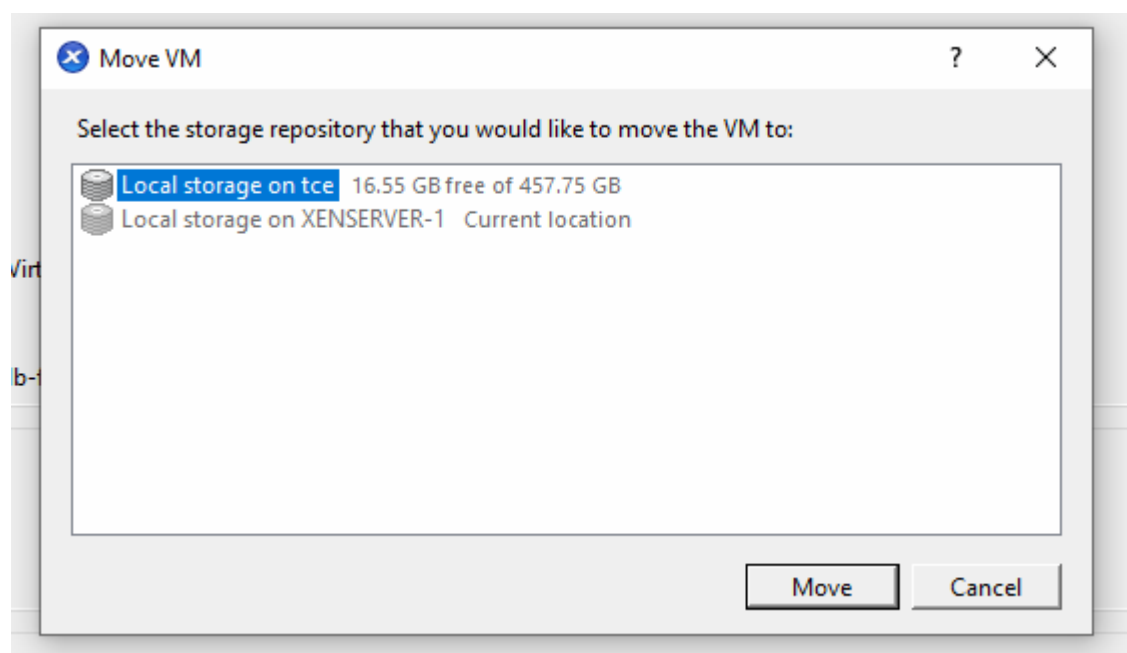


Steps 9:

Right click the vm and select the move vm option from the following

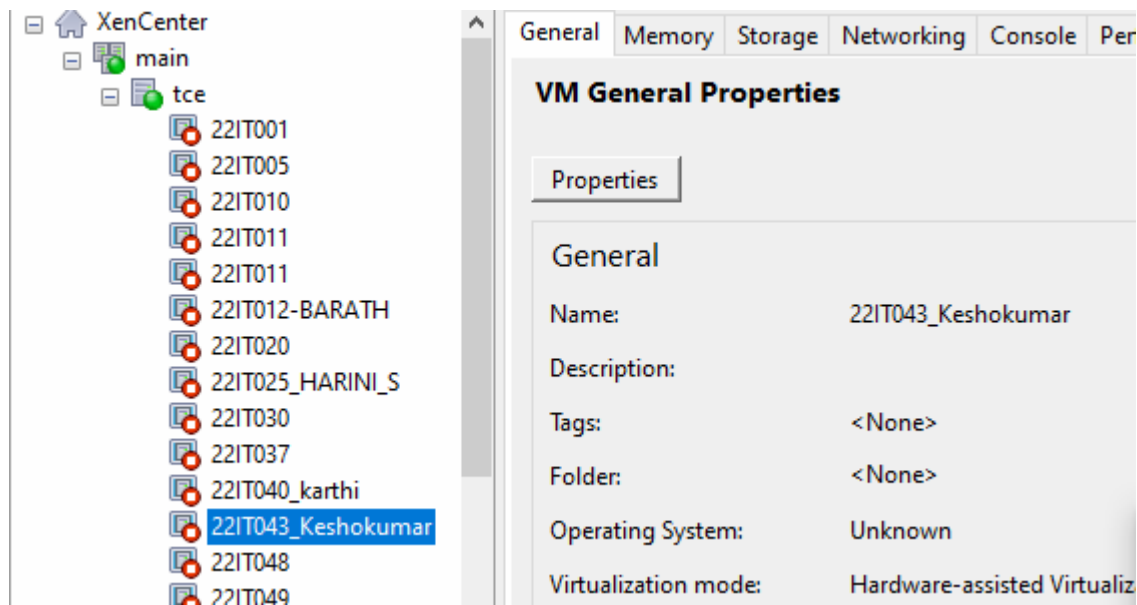


Steps 10: Select the xenserver 2 from the following list



Steps 11:

Now see the Vm in the second server tce and the live migration has been executed successfully.

**RESULT:**

Successfully executed the setup of XenServer and XenCenter; live migration was performed and output verified with zero downtime during VM transfer.